

Paresha Farastu

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SUMMARY

Technical PM, Solutions Architect and published AI researcher (50+ citations, h-index 4) with 5+ years driving product-led growth, conversion optimization, and AI-powered personalization across a 16M+ MAU consumer platform and B2B SaaS spanning housing and healthcare. Track record of running hypothesis-driven experimentation, optimizing end-to-end customer journeys from discovery to conversion, and shipping data-driven features that moved engagement and revenue metrics.

EXPERIENCE

Product Solutions Analyst · EliseAI – CRM Platform (Housing & Healthcare) · NYC, NY Jan 2025 - Present

- Ran discovery and workflow analysis across 100+ enterprise accounts; translated messy, high-volume operational signals into scoped CRM improvements; reduced escalations 30% and improved platform adoption
- Launched enterprise SSO capability; led UAT across Azure AD, Okta, and ADFS, identified critical production issues, including sync strategy bugs and permission inversions
- Mapped full-stack architecture of a multi-repo reporting system spanning Snowflake, Redis, SendGrid, and SQS; surfaced customer-facing defects and translated findings into scoped engineering fixes, reducing support volume by 15x across leasing, collections, and delinquency workflows
- Built Python audit pipeline to systematically detect misconfigured client environments at scale; replaced ad hoc manual QA with a programmatic detection layer, surfacing recurring failure patterns that directly informed product architecture fixes
- Mapped the property go-live pipeline end to end across 30+ PMS integrations (Yardi, Entrata, RealPage, AppFolio, ResMan), CRM guest-card sync, and 15+ calendar providers, converting an undocumented validation flow into a shared triage reference for CS and engineering
- Analyzed failure modes across the onboarding allowlist pipeline; classified step-level failure sub-types to isolate where launches stalled and prioritized recovery automation against the highest frequency blockers
- Scoped event-driven auto-provisioning of sandbox payment infrastructure (Stripe connected accounts, merchant routes, CheckAIt bank chain, GL bank-feed accounts), removing manual seed scripts that blocked end-to-end payment testing; specified idempotency, conflict isolation, and retry vs. dead-letter handling so one bad entity no longer failed an entire batch

Product Manager · Grameenphone Ltd. – MyGP & Easynet · Dhaka, Bangladesh Jan 2019 – Jul 2021

- Owned full product lifecycle across multiple simultaneous features for MyGP (16M+ MAU consumer super-app): defined vision, shaped roadmap, and drove KPIs across acquisition, conversion, engagement, and retention through cross-functional delivery with engineering, design, marketing, legal, and external partners
- Led hypothesis-driven A/B experimentation for News and Shop features; drove 18% engagement growth over 4 quarters through iterative, data-informed product development
- Architected microservices migration for GP Star loyalty platform; redesigned data model, partner integration layer, and pricing engine; +23% retention, 35+ new revenue-generating partners onboarded
- Drove product-led growth via referral campaigns reaching 1M+ new users; applied funnel optimization and SEO/ASO experimentation to improve acquisition and conversion rates
- Led API integrations with third-party partners (e.g., Uber), managing technical requirements, contracts, and go-to-market dependencies across sales and marketing stakeholders
- Managed a 20-person cross-functional team to ship interactive in-app experiences; improved engagement 9% within 2 quarters
- Redesigning Easynet, a zero-rated platform, serving users at their zero moment of truth with data pack offers and partner integrations (Gamebox, O.facebook); revised packaging and onboarding supported 66% revenue growth

Data Analyst · Grameenphone Ltd. – Digital Care · Dhaka, Bangladesh Mar 2018 – Jan 2019

- Designed and prototyped an NLP-powered customer support agent (90%+ accuracy); built real-time data pipelines from 15M+ users using SQL and Python to surface actionable insights and reduce manual resolution time

Software Developer Intern · Hitachi Energy – Velocity Suite · Broomfield, CO May – Dec 2023

- Redesigning a Vendor Registration portal (PostgreSQL, React, Node.js, .NET), improving incident reporting by 47%; developed a CNN classifier on spatial data to identify transmission lines with 65% accuracy

Research & Teaching Assistant · University of Colorado Boulder · Boulder, CO Jan 2022 – May 2024

- Published AI/ML researcher; work on fairness-aware recommender systems using LLMs and multi-agent architectures, in collaboration with [Kiva.org](https://kiva.org) (world's largest crowdfunding platform for social good)

PROJECTS

PromptPulse – Self-Hostable LLMops Platform · Python, Postgres + pgvector, Terraform, Kubernetes

- Built a self-hostable LLMops platform for RAG evaluation CI/CD, prompt drift detection, and retrieval quality monitoring; exposed eval runs and retrieval traces to agentic AI clients over MCP so LLM agents can trigger and interpret evaluations directly. Automated eval pipelines with GitHub Actions and Kubernetes CronJobs, with OpenTelemetry, Prometheus, and Grafana for observability

TikTok Recommendation System · Python, constraint-based optimization, LLM agents

- Recreated TikTok's recommendation mechanisms, using constraint-based optimization for content diversity. Outlined an LLM

based duo-audit moderation system to improve scoring mechanisms, address filter bubbles, and enhance user retention

Multistakeholder Recommendation with Provider Constraints · Gurobi, Python

- Built integer programming models using Gurobi and Python to optimize recommendation systems, improving accuracy by 18% and integrating provider constraints to enhance fairness and user engagement

Inventory Management System for Restaurants · Flask, RabbitMQ, Kubernetes, GCP

- Developed a real-time inventory management system using Flask, and RabbitMQ, featuring automated reordering with notifications, real-time stock tracking, and asynchronous task handling. Implemented RESTful APIs, dockerized the application, and automated deployment and scaling using Kubernetes in GCP

Calendar-Based Productivity App · Flask, SQLite, Heroku, GitHub Actions

- Designed a Flask and SQLite web app for managing tasks and reminders with automatic soccer match scheduling. Deployed on Heroku with continuous integration via GitHub Actions, and implemented unit and integration tests

PUBLICATIONS

- Aird, A., **Farastu, P.**, Sun, J., Štefancová, E., All, C., Volda, A., Mattei, N., Burke, R. "Dynamic Fairness-aware Recommendation Through Multi-agent Social Choice." *ACM Transactions on Recommender Systems*, 3(2), 2025
- Aird, A., All, C., **Farastu, P.**, Štefancová, E., Sun, J., Mattei, N., Burke, R. "Exploring Social Choice Mechanisms for Recommendation Fairness in SCRUF." *FAccTRec Workshop on Responsible Recommendation, RecSys*, 2023 (arXiv:2309.08621)
- Saad, F., Mahmud, H., Kabir, M. R., Shaheen, M. A., **Farastu, P.**, Hasan, M. K. "A Case Study on the Independence of Speech Emotion Recognition in Bangla and English Languages using Language-Independent Prosodic Features." 2021 (arXiv:2111.10776)

AWARDS & HONORS

- **GPSG Senator** · Graduate & Professional Student Government, University of Colorado Boulder
- **Divisional Excellence Award, Grameenphone Ltd. (2020, 2021)** · Awarded twice for the Easynet zero-rated platform
- **Collaboration Award, Grameenphone Ltd. (2020)** · Recognized for cross-functional stakeholder management across engineering, IT, regulatory, InfoSec, SecOps, UX, BI, brand, marketing, pricing, and finance teams

EDUCATION

M.S. Computer Science · University of Colorado Boulder	2024
GPA 3.77	
B.S. Electrical & Electronics Engineering + Physics · BRAC University	2017
GPA 3.83	

CERTIFICATIONS

- **Claude Foundations** · Anthropic
- **AWS Certified Solutions Architect – Associate** · Amazon Web Services (expected Oct 2026)

SKILLS

Product: Full product lifecycle · Roadmapping · PRD writing · Hypothesis-driven experimentation · A/B testing · Funnel optimization · Customer discovery · Market analysis · Pricing & packaging · Value proposition · Product-led growth · Backlog prioritization · Stakeholder alignment · Customer Success collaboration · Enterprise API integration · Webhook architecture · OAuth 2.0 · SSO · SAML · SCIM · RBAC · User provisioning · Audit logging · Technical discovery · PoC development · Enterprise onboarding · Pre-sales support · Production debugging

AI/ML: LLM application development · Prompt engineering · RAG · Model Context Protocol (MCP) · Agentic workflows · Eval design · Recommender systems · Fairness-aware ML

Tools: Mixpanel · Firebase · Google Analytics · Datadog · Power BI · JIRA · Figma · Zendesk · Retool · LogRocket

Technical: Python · SQL · JavaScript · React.js · Node.js · PostgreSQL · Snowflake · Fivetran · GCP · AWS · Docker · Kubernetes · RabbitMQ